

Living vs non-living

Definition:

- **Living things** grow, reproduce, use energy, respond to the environment, and are made of cells. **Non-living things** do not do these things.

Explanation:

- Living things show life processes: they eat or make food, breathe or exchange gases, grow, move (some slowly), and can make more of their kind. Non-living things (rocks, toys) don't grow, don't need food, and don't reproduce.

Examples:

- Living: a dog, a tree, bacteria (tiny). Non-living: a chair, a cup, the sun (note: the sun is non-living but causes life processes on Earth).

Activity:

- Put pictures on cards and have children sort into "Living" / "Non-living" and explain why (look for growth, breathing, eating).
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Needs of living things (food, water, air)

Definition:

- Living things need certain things to survive; the three basics are food (energy), water, and air (oxygen or carbon dioxide for plants).

Explanation:

- Food gives energy and building materials. Water is used for chemical reactions and transport inside bodies or plants. Animals use oxygen to release energy from food; plants use carbon dioxide, water, and light to make food. Without these, living things cannot grow or survive.

Examples:

- Plants: water + light + CO₂ → food. Animals: eat plants/other animals and breathe air.

Activity:

- Grow two beans: one watered and one not, observe differences. Ask children which needs were missing.
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Plants and animals differences

Definition:

- Plants usually make their own food and have roots/stems/leaves; animals move and eat other organisms for food.

Explanation:

- Plants have parts for making and transporting food (leaves, vessels) and usually stay in one place. Animals often move to find food, have sense organs, and eat plants or other animals. There are exceptions (some plants move slightly; some animals are sessile as adults), but these differences help young learners group organisms.

Examples:

- Plant: sunflower (makes food by photosynthesis). Animal: cat (eats other organisms).

Activity:

- Create a two-column chart and let students place pictures under “Plant” or “Animal,” then ask them why they placed each there (food source, movement).